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Wildfire Early-Detection System

What?

This project is a dual-Arduino prototype designed to **monitor** temperature, humidity, gas concentration, and water presence for early wildfire detection. It functions as a low-cost **IoT device** for on-site logging and **BLE transmission** for real time monitoring.

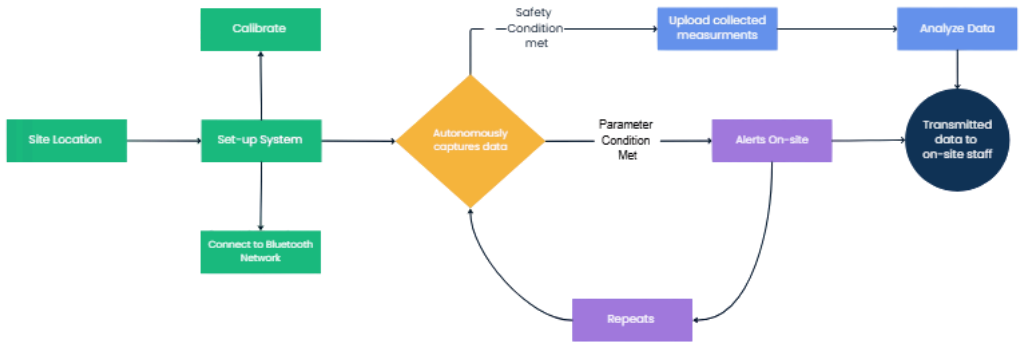
How?

- **Design:** Created a compact enclosure in **SolidWorks**, for component clearance.
- **Integration:** Utilized a **two-microcontroller transmission** with both boards handling different hardware components.
- **Programming:** Used **Arduino IDE** to **calibrate** sensor readings, communication paths, and datalogging.
- **Fabrication: Incorporated additive** manufacturing to 3D print enclosure while soldering and housing electrical system.

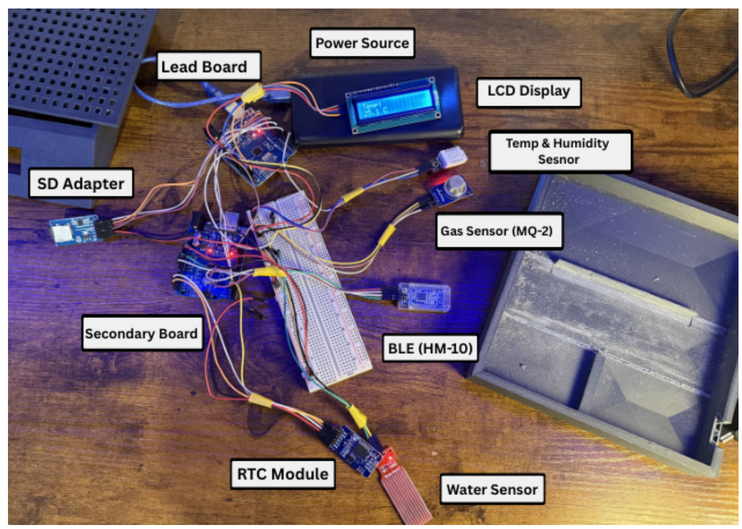
Results?

- Achieved data collection across all sensors with synchronized time logging
- Established **Bluetooth range of 40 meters,** maintain consistent communication without loss.
- Enabled **tracking and analysis** of transmitted data for identifying visual environmental trends and early hazards indicators
- Maintained total prototype **cost under $50** for reliable system

System Process



Electrical System



Final Form

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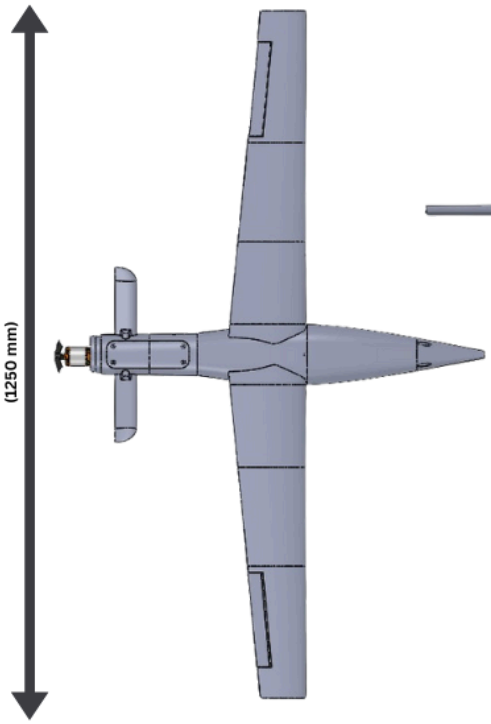
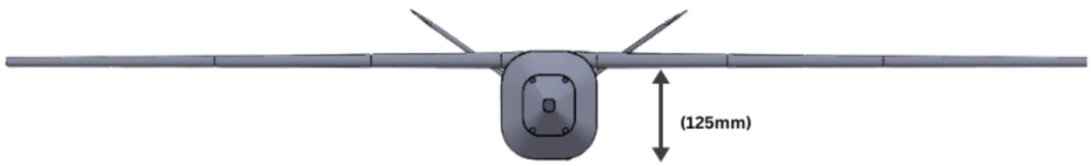
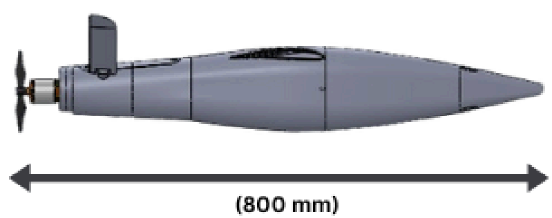
Deployable Aerial Reconnaissance Vehicle (DAR-V)







What?

Developed as a small team of engineering students, the Deployable Aerial Reconnaissance Vehicle (DARV) is a fixed-wing UAV designed for autonomous flight, aerial reconnaissance, and environmental monitoring.

I led the avionics integration, overseeing the setup of the flight controller, GPS, and telemetry communication systems to ensure stable and coordinated operation.

The design goal was to achieve a sustainable flight time of 20 minutes within a 3-mile operational radius, using 3D- printed modular airframe optimized for lightweight performance and field serviceability.

How?

- Airfoil & Structure: Selected the Eppler 193 airfoil through Computational Fluid Dynamics (CFD) analysis for it’s high lift-to-drag ratio and stable low-speed performance at 40 mph.
- Design: Modeled the full airframe and internal mounting layout in SolidWorks, incorporating sections for the flight controller, GPS, video transmitter, receiver, and GPS module.
- Integration: Configured INAV flight controller program, mapping servo outputs and motor mixing for coordinated pitch, roll, and yaw control.
- Manufacturing: Employed Fused Deposition Modeling (FDM) 3D printing with primary focus on print orientation, wall thickness, and infill pattern to enhance the mechanical strength and reduce deformation under aerodynamic loading

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Results (In-Progress)

- Structural Failure Analysis: Using SolidWorks, we identified the primary structural weakness at the epoxy-bonded joint connecting the side fins to the central fuselage. Under aerodynamic loading, the joint became a failure point leading to crashes. A case study incorporating distributed weight, center of gravity, and central valley conditions confirmed our prior assumptions of malfunction.

- Flight Stability: After multiple test flights, the UAV exhibited instability and structural integrity issues, particularly at the rear fins and fuselage joint. This led to a redesign of the rear control finds, replacing the original short and long linkage system that drove the servos with a more rigid, direct mount configuration to improve reliability and response.

- Accessibility Issues: Leading into set up processes for flight testing, we experienced difficulty accessing the internal avionics system. Often requiring removal of multiple fasteners which resulted in component overheating during extended testing. To resolve this, we redesigned the front cone compartment with a drop-out access method, allowing rapid swaps and debugging without full disassembly of the airframe.

Analysis

After our flight tests, we found that the weak points of structural integrity were at the epoxy joints.

Another primary issue was the dispatchment of the motor mid-flight leading to loss of control and crash. Leaving the team with more redesign protocol.

Initial Form

Final Form

Flight Test #2

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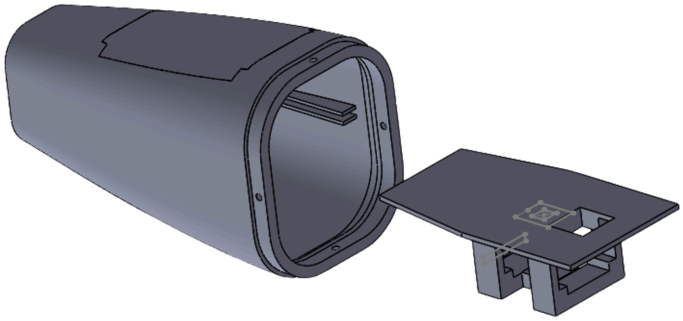
Systems Redesign Overview

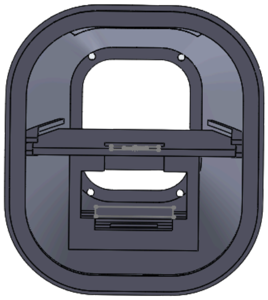
Avionics Compartment

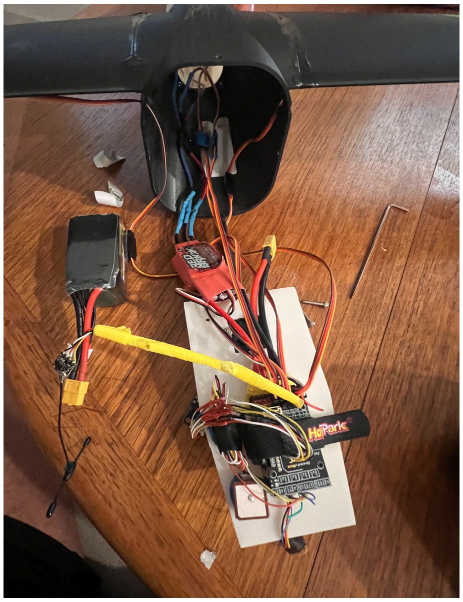


Purpose: Serves as the internal electrical system within aircrafts to manage the UAV flight control and navigation, linking sensors, servos, and telemetry modules to maintain stable and responsive operation during flight

Key Components:

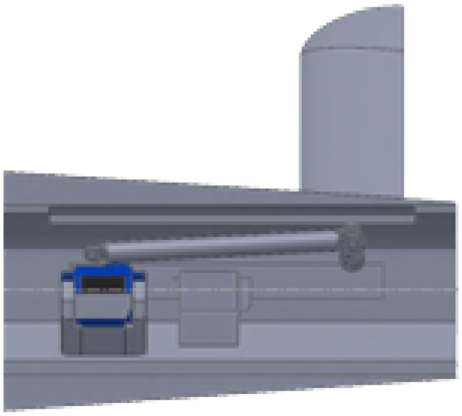
- **Flight Controller:** Brain of the UAV, interpreting sensor data and adjust motor outputs
- **GPS module:** Provides position parameter for navigation & tracking
- **Telemetry Module:** Sends real-time flight data
- **Receiver & Transmitter:** Handles communication between controller the UAV
- **Electronic Speed Controller (ESC):** Regulates speed of motor rotations
- **FPV Camera & VTX:** Provides first person video feedback during flight



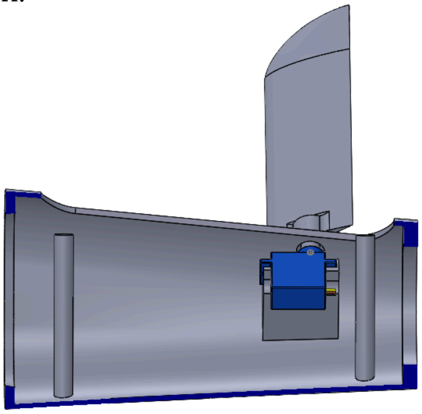


Rear Wing Fuselage

Initial Design: Original rear-fin actuation used offset linkages due to space constraints; testing exposed friction between the links and servos, giving us inconsistent control



Redesign: The rear fins were redesigned for direct servo actuation, placing both servos side by side with a roller bearing in-between to reduce friction and improve control precision.



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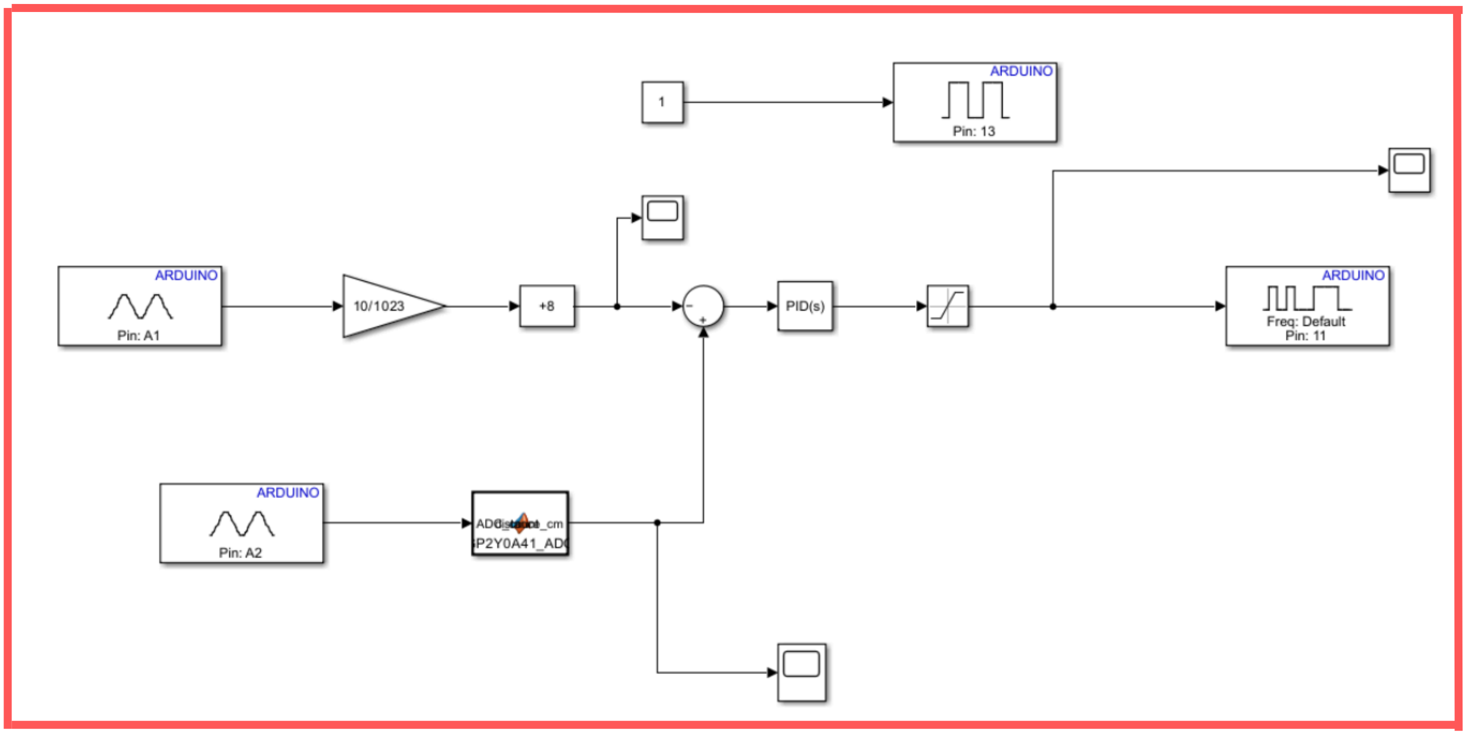
Ball-Levitation Control System

Simulink Model



What?

This project is a closed-loop ball levitation system that uses an IR distance sensor and PWM controlled blower to hold a ping-pong ball at a set height inside a tube. Build using Simulink model based design with real-time embedded deployment to an Arduino for dynamic airflow control.

How?

- **Modeling:** Developed a **Simulink** control converting nonlinear **IR sensor** voltage into **position feedback** for closed-loop regulation.
- **Control:** Implemented a **PID** based controller to modulate the wind blower **PWM** and stabilize the ball at various setpoints.
- **Integration:** Interfaced the IR sensor, motor driver, and blower through an Arduino, **deploying** the Simulink model for **real-time** execution.
- **Testing:** Tuned gains, **filtered noise,** and **refined logic** to maintain stable levitation

Diagram setup

- Top strand: Sends a constant value of 1 to Digital pin to keep the blower motor activated.
- Middle strand: Converts the potentiometer's heigh input into a velocity command with an 8 cm added offset for sensor correction.
- Bottom strand: Uses MATLAB function to turn the IR sensor voltage into a distance reading.
- Sum block: Compares setpoint and sensor distance to generate control error for airflow.

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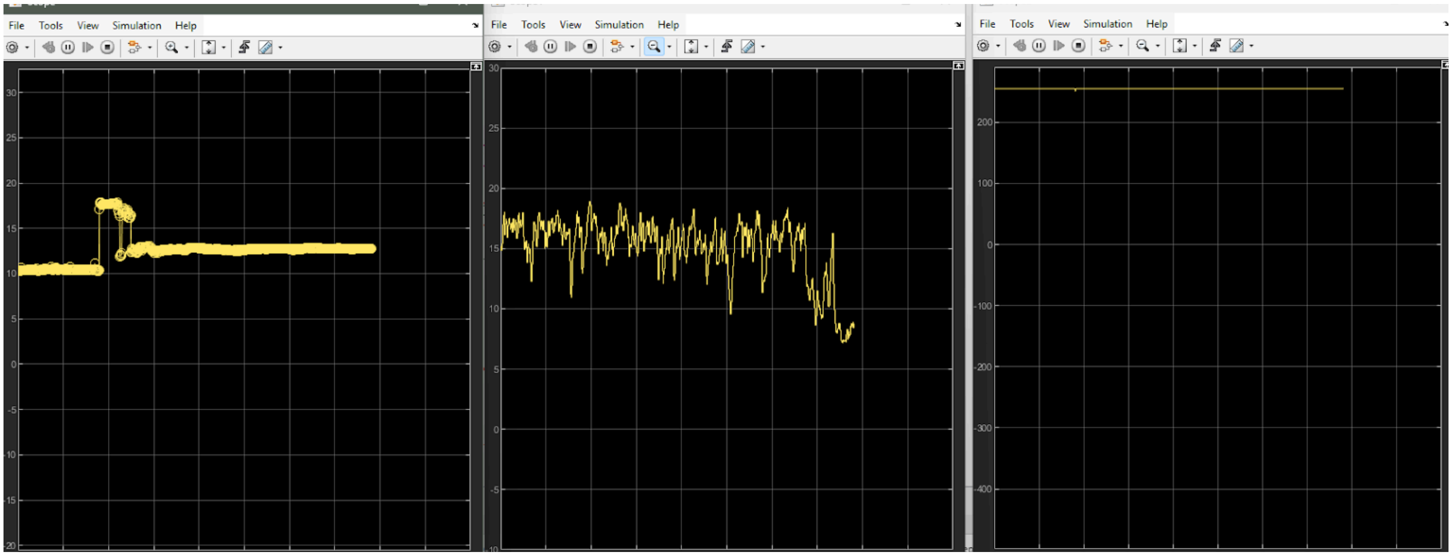
Results?

- Achieved stable leviation across multiple setpoints with real-time control.
- Validated distance conversion and reliable sensor feedback.
- Demonstrated effective closed-loop airflow regulation using Simulink deployed embedded control.
- Worked on PID tuning, sensor integration, and model driven development.

Test #1: Reference pt. set at 13cm

Analysis

The scope results show the system's behavior when the potentiometer was adjusted to a **13cm** reference height. The **left graph** confirms the setpoint transition as the **potentiometer** moves and adjust to the desired reference. The **middle graph** shows the **IR sensor** tracking the ball's real time position, hovering near the **13cm level** with natural fluctuation from airflow dynamics. The **right graph**, represent **Pulse Width Module (PWM) output,** remains steady indicating that the system reached a point of stabilization. Together , the graphs demonstrate that the system was consistent for the intended application.



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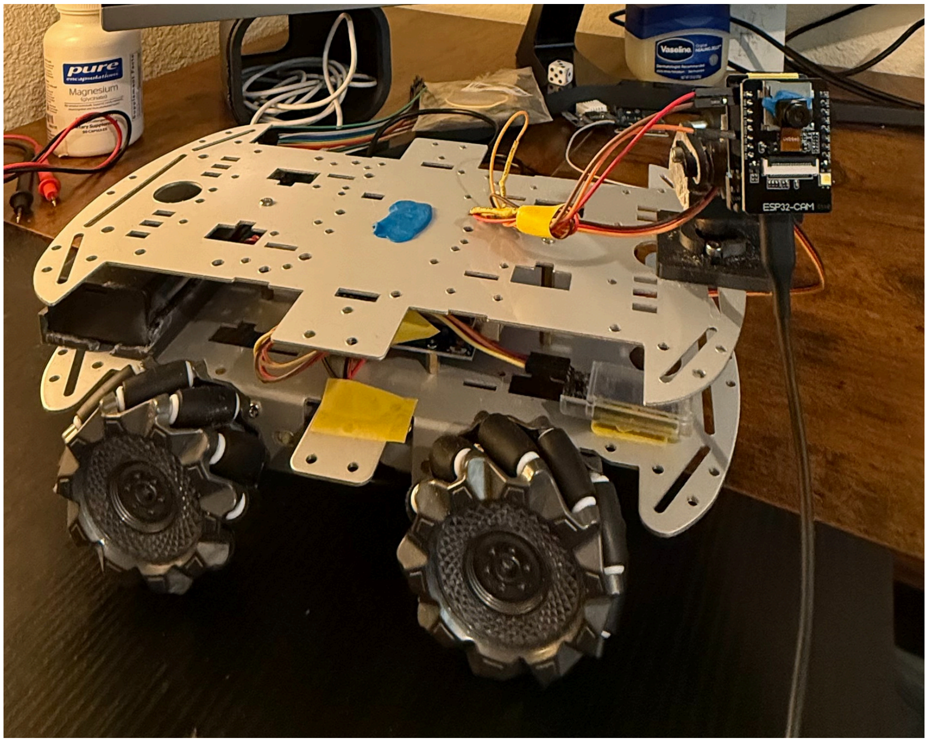
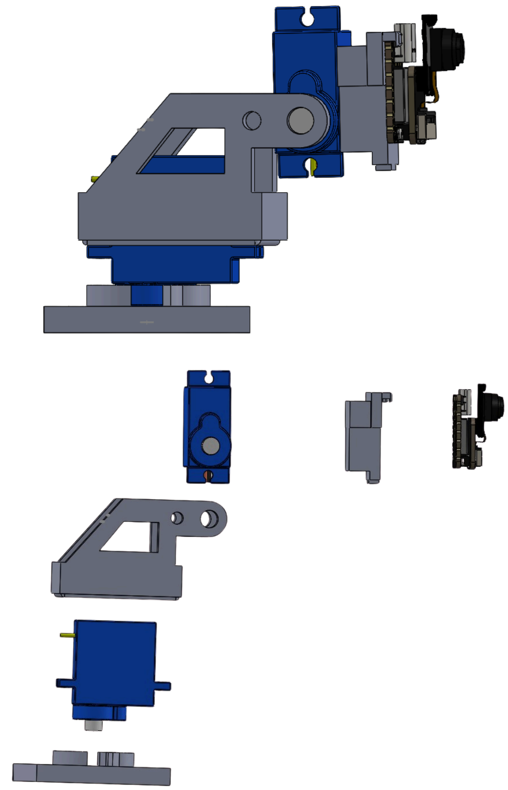
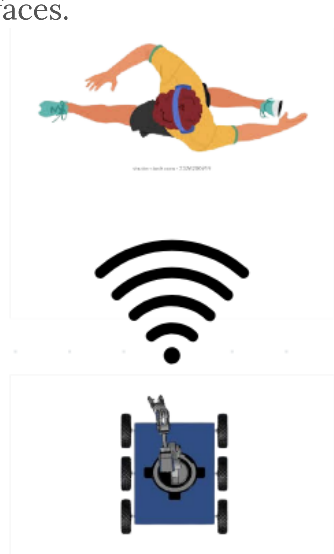
Surveillance and Control Autonomous Rover (SCAR)





What?

This Project is a six-direction rover designed to integrate camera motion tracking, omnidirectional movement, and wireless video transmission. That can be third party streamed to web server surfaces.



How?

- **Design:** Created a two degree of freedom (2DOF) servo mount in SolidWorks giving control for Pan and tilt for the camera system.
- **Integration:** Incorporated an ESP32-CAM module for live streaming, motion-detection triggered recording, and SD card storage. Added an HM-10 BLE module for for mobile surface control.
- **Programming:** Developed systemic code in Arduino IDE to produce an URL website for First Person Video (POV).
- **Fabrication:** Designed and 3D printed servo mount for camera module to ensure stable recording.

Results

Was able to construct s reliable omnidirectional controlled rover vehicle capable of capturing video regulation, utilizing probe efficiencies, with secure data logging accessible from 50 feet out.

This project reinforced skills in mechanical design, introduction into IoT communication, and embedded system programming into one autonomous and responsive platform.

A practical application for this rover includes remote surveillance. Enabling safer understating of dangerous environments without direct human involvement.